microbiota, but these modifications occur at the level of individual strains and species and are not easily predicted a prior.

7.3.3 Synbiotic

In 1995, Gibson and Roberfroid introduced the term "synbiotic" to describe a combination of synergistically acting probiotics and prebiotics. As the word "synbiotic" implies synergy, the term should be reserved for those products in which a prebiotic component selectively favours a probiotic microorganism. Synbiotics have both probiotic and prebiotic properties and were created in order to overcome some possible difficulties in the survival of probiotics in the gastrointestinal tract.

7.3.4 Phytochemicals

Phytochemicals are plant chemicals that differ from nutrients in some important ways. Phyto is a greek word for plants. Essential nutrient which include protein, fats, minerals, and vitamins are essential for life. Phytochemicals are not necessary for life but they help to promote optimal health by lowering risk for chronic diseases, such as cancer and heart disease. They are found only in plant foods. Fruits and vegetables are among the best sources of these compounds. Phytochemicals are believed to have many health benefits and prevent lifestyle diseases.. Some groups of phytochemicals have been linked to decreased cancer risk also. Following are examples of some phyto chemicals with nutritional importance.

7.3.4a Flavanoids

Flavanoids are a special class of phytochemicals that includes hundreds of different compounds. They are excellent antioxidants and some have hormonal properties. Among some of the most studied flavonoids are allicin, which is found in onions and garlic.

Benefits of Flavanoids

1. Longer life span
2. Prevents obesity and helps in weight management
3. Prevents cardio vascular disease, diabetes, cancer.
4. Prevents neuro generative disease
5. Slows down ageing process.

7.3.4b Carotenoids

Carotenoids are a group of phytochemicals that act as pigments, giving plants their bright green, orange, yellow, red, and blue colors.

Benefits of carotenoids

- Beta-carotene, found in carrots, sweet potatoes, green leafy vegetables, red peppers, and pumpkin. Beta-carotene from foods has been linked to a reduced risk for lung cancer.
- Lycopene, found in tomatoes and strongly linked to reduced risk for prostate cancer.
- Lutein, found in green leafy vegetables and linked to reduced risk for cancer and macular degeneration.

7.3.4c Antioxidants

Antioxidants are our first line of defense against free radical damage, and are critical for maintaining optimum health and wellbeing. Antioxidants are carotenoids, lycopene, vitamin C, vitamin A, vitamin E etc..

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